

Design Exercise 1

02/06/25 - sketching initial specs and design

Eng Notebook - Savanna Coffee

goals: need to be able to:

- create account
- log in to account
- list accounts via text wildcard - so if I needed to search for an account, I can
- send message to a recipient through the central server
- read messages: if someone is logged in, deliver immediately; else keep a queue
- delete messages
- delete accounts

MY ASSUMPTIONS

- all clients and the server are written in Python
- if a client sends a message to another client, that message goes through the central server, which then forwards it to the client
- our passwords have to be hashed. We will do this on the client side and only send hashed passwords over the wire
- clients cannot send messages to each other until they have established a connection with the server
- store unread messages in a priority queue - likely array in Python with most recent message last in the queue to preserve order
- when we delete an account, if it has unread messages, we prompt if the user wants to read them. they can say yes or no
- we send over all messages as utf-8 encoded strings. the strings contain sender, receiver, message info in the form:

{ uuid | datetime | sender | receiver | message content }

↳ this is a dict that gets turned into a string → encoded in binary and sent over the network

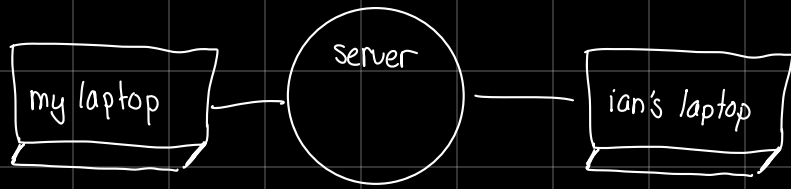
↳ messages saved to internal database as the dict, so the server can do operations on it

Wire Protocol

02/07/25 - 02/08/25

→ going to follow TCP/IP

→ send encoded strings over the network under utf-8



→ need to check multiple sockets at once to read/write

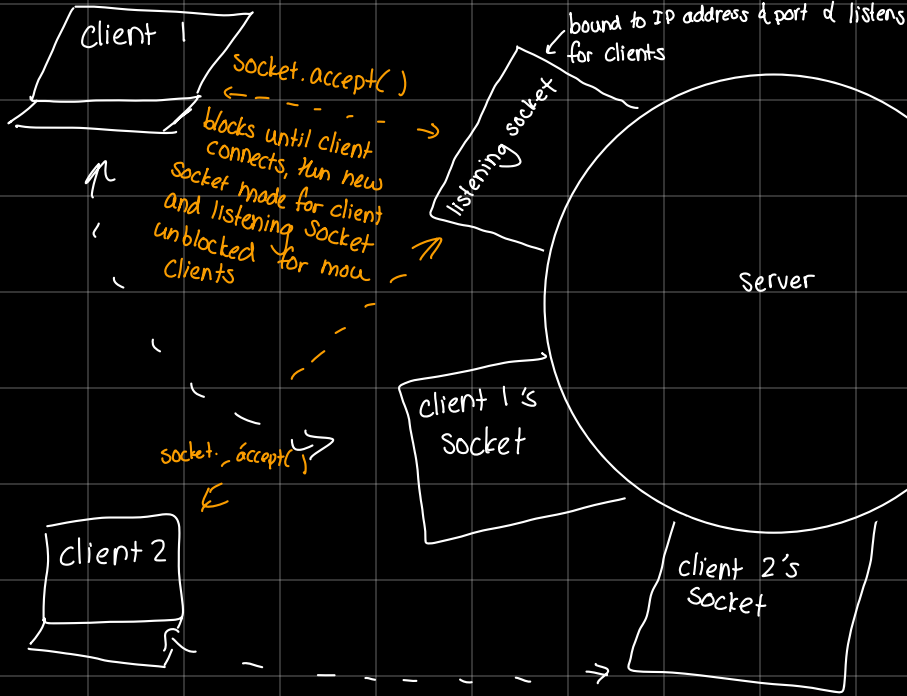
to/from pipe w/ client

→ `Select.select()` checks multiple sockets at once

checks read, write, exception list of sockets

• listening socket set to `socket.setblocking(False)`

so listening socket continuously polls instead of blocks



Direct messaging system

→ server stores a map of usernames to client sockets

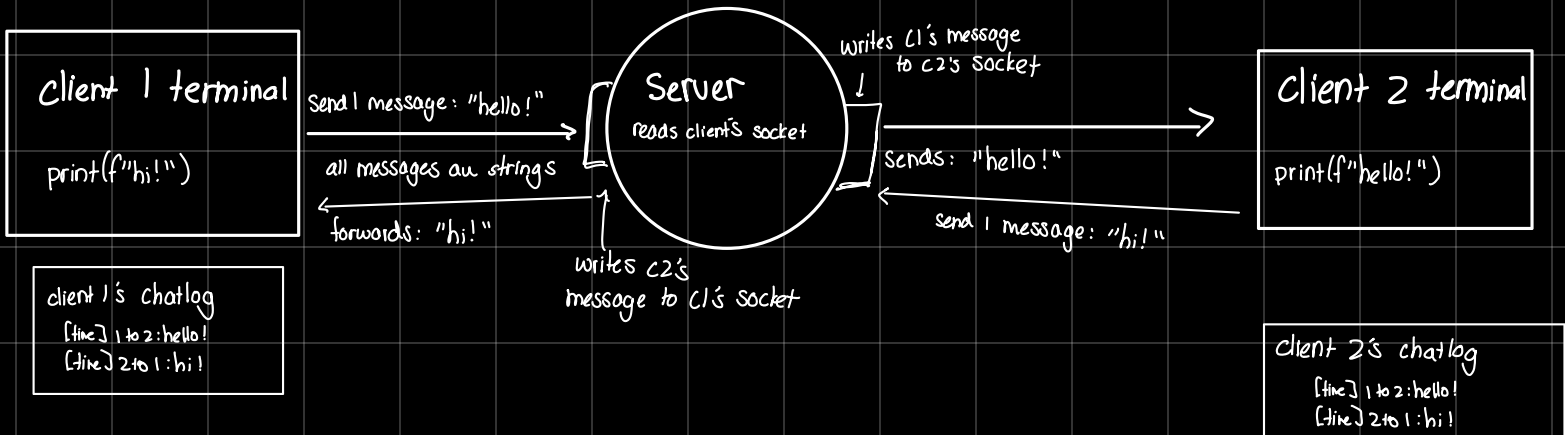
↳ server routes to specific client

→ client specifies which username to send the message to

- server will pass messages between clients by looping over `Select()` calls

- chatlogs get updated w/ timestamps, user info every time a message is sent or received

→ server stores map of clients to sockets by:
{username, socket}, draws from receiver key in message to find the right socket



Wire Protocol: We should be able to switch between JSON mode & normal mode by flipping a boolean.

→ Our custom wire protocol sends dicts that are "stringified" into strings encoded as utf-8. Ex: `{ "uid": 1, "datetime": "2025-07-02T12:00:00", "sender": "client1", "receiver": "client2", "content": "hello!" }`

↳ strings get sent over the network as binary using utf-8 conversion

Creating a new account:

02/07/25 - Working on data management and networking

- when we log into an account, establish the client-server connection via `accept()` & other calls (easy)

- then an exactly as many sockets as there are currently logged-in users

- server maps: $\{ \text{username} : \text{assigned socket} \}$

- Hash password client side

```
accounts = {  
  uuid: 1 | user: _____ | pass: _____ }  
  uuid: 2 | ...  
  uuid: 3 | ...  
}
```

Offline messaging:

02/08/25: Messaging - Harder than I thought!

- if the desired client is offline, the server will store the messages in a dictionary with the same format as the other messages

```
pending_messages = {  
  'username 1': [ "this is a message you'll read later!" ]  
  'username 2': [ "this is another, different message!" ]  
}
```

- when user logs back in, parse dictionary for a key match - if one exists, prompt that client & display # of messages

if client wants to see it, deliver messages by writing to client's socket

Clients Storing Chatlog and Deleting select messages

02/09/25

- read messages and messages client x writes will be stored in client x's chatlog

02/09/25

- debating whether to use plaintext or something else - if we write to log file we should be able to see everything (update: yes on .txt)

- if client x deletes a message from client y, it does not show up for client y as a deletion (so only receiver can delete messages)

chatlog

[timestamp] - alice to bob: "hello!"

[timestamp] - bob to alice: "hi!"

[timestamp] - bob to alice: "good morning"

- delete messages by parsing through

log by message content, then
remove that line from plaintext

- once deleted, gone from log

- if queued messages read, they are immediately
added to the chatlog

chatlog

[timestamp] - alice to bob: "hello!"

[timestamp] - bob to alice: "good morning"

-if an account has unread messages, prompt user if they want to read them before deletion. if no, server deletes them

-should the server log everyone's full chatlog? or only unread?

Server Stores

→ dict of all registered usernames & hashed passwords (enforce uniqueness in usernames & passwords - no hash collisions!)

→ unread messages to offline clients

→ full chatlogs (with deletions)

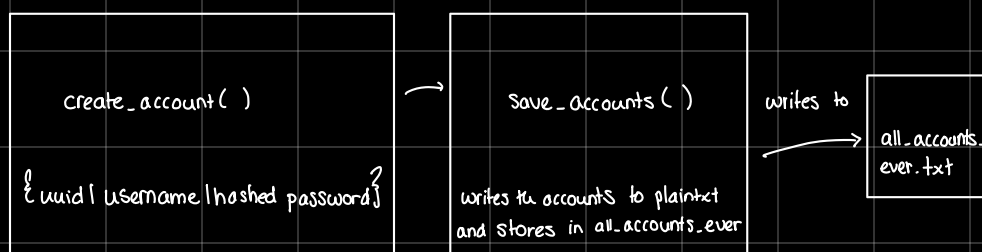
→ keeps track of who is messaging who. its in our interest to make this as simple as we can

keep client minimal: server should be universal source of truth. Clients shouldn't store persistent data and should only talk to the server and return values for the GUI to display.

Misc

→ messages in real time are displayed in the terminal when received ✓

→ do we want to prompt the client every time and ask if they want to see the message? Could get annoying



02/10/15: Working on GUI. I've never done GUI stuff before so I'm nervous. (Reading up on Tkinter & writing it today probably)

-Update: This was the most annoying part ever. I shouldn't have done this at the end.

- I wrote a script that simulates client/server communications using our wire protocol on test sockets! (adding it to testing suite)
- Results were interesting:

Test Summary

4 tests run in total.
All tests passed!

Detailed Byte Size Totals (Application Payload Only):

Custom (Plain) Mode:

Sent: 22 bytes
Received: 22 bytes
Total: 44 bytes

JSON Mode:

Sent: 41 bytes
Received: 49 bytes
Total: 90 bytes

- For both sending and receiving data, it looks like we're outperforming JSON by close to 50%. Since our custom wire protocol only requires sending two pieces of data (recipient, message) for sending, and only the message contents themselves upon receiving, we're already sending over a lot less raw content than the JSON approach.
- Also, JSON requires serializing extra characters like braces and dict separation info. We are literally just sending over strings with occasional special characters, such as sending recipient and message info.
- Critically, data isn't parsed into a dict until after it hits the server. This means we're storing heavy info chunks, but sending the bare minimum over the server.
- We could even compress our protocol further using zlib, which would make us even more compact.
- TCP/IP overhead is about the same for both methods. ~140 bytes.
- Also interesting is that JSON has different byte amounts for sending & receiving, unlike us. It seems like the server adds a few extra bytes when it sends messages to the client. Our implementation is much more standardized in this respect!

